

&" NuclearCraft

Plutonium Age

v1.0.0 Production — Complete Setup & Command Guide

Paper / Purpur 1.21+ · Java 21+ · SQLite / MySQL

- ' 14 development phases — fully complete and production ready
- ' Radiation system with 5 escalating stages and contagion spread
- ' Nuclear Smelter, Forge & Titan Reactor Forge machines
- ' Plutonium Titan multi-phase boss encounter (up to 50+ players)
- ' Irradiated zombies, mutated crops, custom ores & combat mastery
- ' Full admin suite: health checks, diagnostics, memory monitoring, data repair
- ' SQLite (zero config) or MySQL for high player count servers
- ' Resource pack: custom 3D models + 30 custom sound events

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1. Installation & First Boot

Server Requirements

Requirement	Details
Server software	Paper 1.21+ or Purpur 1.21+ (1.21.4 recommended)
Java version	Java 21 or newer (Temurin / OpenJDK 21)
RAM	Minimum 2 GB allocated (-Xmx2G), 4 GB+ recommended for SMP
Database	SQLite (built in, zero config) or MySQL / MariaDB for 50+ players
Permissions	LuckPerms recommended, any Bukkit-compatible plugin works

Step-by-Step Installation

1	Download Get NuclearCraft-1.0.0.jar from your release package.
2	Place JAR Copy NuclearCraft-1.0.0.jar into your server's /plugins/ folder.
3	First boot Start the server. NuclearCraft auto-generates all config files. Look for the green startup banner in console.
4	Verify Run /nc version in-game or console. Expect: "NuclearCraft: Plutonium Age v1.0.0 Production"
5	Configure Edit files in plugins/NuclearCraft/ to tune radiation, machines, boss and balance. Run /nc reload to apply.
6	Resource pack Host the .zip at a public HTTPS URL and configure resourcepack.yml (see Section 3).

Generated Folder Structure

plugins/NuclearCraft/	
%%% config.yml	Main: database, performance, worlds
%%% radiation.yml	Stages, damage, decay, contagion
%%% titan.yml	Boss HP phases, ability timing
%%% titan_items.yml	Boss loot table
%%% smelter.yml	Recipes, fuel, overheat
%%% forge.yml	Upgrade tiers, energy cost
%%% farming.yml	Crop growth, antidote recipes
%%% zombies.yml	Spawn rates, surge settings
%%% ore.yml	Vein size, depth, rarity
%%% combat.yml	Weapon mastery, damage modifiers
%%% equipment.yml	Plutonium / Hazmat armor stats
%%% messages.yml	All player-visible strings
%%% resourcepack.yml	Pack URL, hash, prompt
%%% balance.yml	Economy rewards, XP rates
%%% dumps/	Created by /nc dumpdata

2. Database & Performance Config

SQLite (Default — Zero Config)

SQLite is enabled by default. The database file (nuclearcraft.db) is created automatically in the plugin folder. No changes needed for servers up to ~50 players.

MySQL / MariaDB Setup

Switch to MySQL for larger SMP servers (50+ concurrent players) for better concurrent write throughput.

```
# plugins/NuclearCraft/config.yml
database:
  type: mysql          # Change from "sqlite"
  host: "127.0.0.1"
  port: 3306
  name: "nuclearcraft"
  username: "nc_user"
  password: "your_password"
  pool-size: 10
  connection-timeout: 30000
```

MySQL setup SQL:

```
CREATE DATABASE nuclearcraft CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'nc_user'@'localhost' IDENTIFIED BY 'your_password';
GRANT ALL PRIVILEGES ON nuclearcraft.* TO 'nc_user'@'localhost';
FLUSH PRIVILEGES;
```

Performance Settings

```
# config.yml
performance:
  monitor-interval-seconds: 30
  log-stats: false
  particle-throttle-percent: 50  # Reduce particles below this TPS

admin:
  enable-memory-sampler: true
  memory-warn-threshold-percent: 85
  dump-dir: "dumps"
```

World Whitelist

NuclearCraft is active in all worlds by default. Restrict it with:

```
# config.yml
worlds:
  whitelist-enabled: false
  whitelist:
    - world
    - world_nether
    - world_the_end
```

Recommended JVM Flags

```
java -Xms2G -Xmx4G \
```

```
-XX:+UseG1GC -XX:+ParallelRefProcEnabled \  
-XX:MaxGCPauseMillis=200 -XX:+UnlockExperimentalVMOptions \  
-XX:+DisableExplicitGC -XX:+AlwaysPreTouch \  
-XX:G1NewSizePercent=30 -XX:G1MaxNewSizePercent=40 \  
-XX:G1HeapRegionSize=8M -XX:G1ReservePercent=20 \  
-XX:InitiatingHeapOccupancyPercent=15 \  
-Daikars.new.flags=true \  
-jar paper.jar --nogui
```

3. Resource Pack Setup

What the Resource Pack Provides

- Custom 3D item models for all Plutonium and Titan equipment
- Custom models for Nuclear Smelter, Forge, and Titan Reactor Forge
- 30 unique sound events: radiation, machines, boss, combat
- Visual identifiers to distinguish NuclearCraft items from vanilla

Step 1 — Host the ZIP at a Public HTTPS URL

Host Option	How
GitHub Releases	Upload to a GitHub release, copy the "Download" link directly.
Dropbox	Upload and share. Change ?dl=0 to ?dl=1 in the link.
Your Web Server	Serve via nginx/Apache with HTTPS. Any static file host works.
Modrinth CDN	Upload as a project file and use their CDN download URL.

Step 2 — Generate the SHA-1 Hash

The hash allows clients to skip re-downloading an unchanged pack.

```
# Linux / macOS
sha1sum "NuclearCraft Resource Pack 1.0.1.zip"

# Windows (PowerShell)
Get-FileHash "NuclearCraft Resource Pack 1.0.1.zip" -Algorithm SHA1

# Output example (40 hex characters):
# a3f2c1d4e5b6a7c8d9e0f1a2b3c4d5e6f7a8b9c0
```

Step 3 — Configure resourcepack.yml

```
# plugins/NuclearCraft/resourcepack.yml
resource-pack:
  enabled: true
  url: "https://yourdomain.com/NuclearCraft-ResourcePack-v1.0.0.zip"
  hash: "a3f2c1d4e5b6a7c8d9e0f1a2b3c4d5e6f7a8b9c0" # your actual hash
  required: false # set true to kick players who decline
  prompt: "NuclearCraft requires a resource pack for custom models and sounds."
  log-status: true
```

Step 4 — Reload

Run /nc reload. All new players joining will automatically receive the pack prompt. Existing online players must relog to receive it.

TIP: Set required: true on competitive SMP servers so every player sees correct item models. Players who decline on required: false will see vanilla placeholders instead of NuclearCraft 3D models.

Sound Events Reference

These 30 sound keys must be declared in your resource pack's sounds.json:

nuclearcraft.radiation.gain	nuclearcraft.radiation.stage_change
nuclearcraft.radiation.cure	nuclearcraft.radiation.surge
nuclearcraft.radiation.death	nuclearcraft.item.antidote_drink
nuclearcraft.item.serum_drink	nuclearcraft.ore.discover
nuclearcraft.ore.mine	nuclearcraft.ore.drill
nuclearcraft.machine.smelter_start	nuclearcraft.machine.smelter_complete
nuclearcraft.machine.smelter_overheat	nuclearcraft.machine.forge_start
nuclearcraft.machine.forge_complete	nuclearcraft.machine.forge_overload
nuclearcraft.machine.titan_forge_start	nuclearcraft.machine.titan_forge_complete
nuclearcraft.upgrade.success	nuclearcraft.upgrade.fail
nuclearcraft.combat.radiation_hit	nuclearcraft.combat.aura_pulse
nuclearcraft.titan.spawn	nuclearcraft.titan.death
nuclearcraft.titan.phase_change	nuclearcraft.titan.roar
nuclearcraft.environment.radiation_night_start	nuclearcraft.environment.radiation_night_end
nuclearcraft.environment.radiation_cloud	

4. Permissions Reference

NuclearCraft uses a hierarchical tree. nuclearcraft.admin grants all sub-nodes automatically.

Permission Node	Default	Description
nuclearcraft.use	all	Use /nc radiation check on yourself
nuclearcraft.reload	op	/nc reload — reload all configs
nuclearcraft.debug	op	/nc debug — toggle debug logging
nuclearcraft.give	op	/nc give — give custom items
nuclearcraft.admin	op	Full admin access (grants all children below)
nuclearcraft.admin.radiation	op	Radiation admin: add/remove/set/clear/check others
nuclearcraft.admin.zombies	op	Zombie admin: spawn types, surge control
nuclearcraft.admin.ore	op	Ore admin: spawn blocks, give drill/fragments
nuclearcraft.admin.smelter	op	Smelter admin: give, stats, debug
nuclearcraft.admin.equipment	op	Equipment admin: give plutonium/hazmat gear
nuclearcraft.admin.farming	op	Farming admin: give seeds/cures, growall
nuclearcraft.admin.forge	op	Forge admin: give, energy, tier upgrade
nuclearcraft.admin.combat	op	Combat admin: stats, mastery, kills
nuclearcraft.admin.titan	op	Titan admin: spawn, kill, phase, stats + Titan Tech
nuclearcraft.admin.debug	op	/nc debug <system> diagnostic subcommands

LuckPerms Setup

```
# Grant full admin to the admin group:
lp group admin permission set nuclearcraft.admin true

# Grant basic use to all players (default group):
lp group default permission set nuclearcraft.use true

# Grant a single specific node:
lp user Steve permission set nuclearcraft.admin.titan true
```

5. Commands — Core & Debug

All commands support the alias `/nc` (short for `/nuclearcraft`). Commands shown below work in-game and in the server console (console omits the `/`). Tab completion is available for all commands.

Core Commands

`/nc help`

Show the help menu filtered to your permissions

Permission: nuclearcraft.use

`/nc version`

Display plugin version, Java version, server platform

Permission: nuclearcraft.use

`/nc info`

Show runtime counts: ores, machines, crops, players, boss status

Permission: nuclearcraft.use

`/nc reload`

Reload all config files without restarting the server

Permission: nuclearcraft.reload

`/nc give <player> <item> [n]`

Give a custom NuclearCraft item to any player

Permission: nuclearcraft.give

`/nc debug`

Toggle debug logging on/off

Permission: nuclearcraft.debug

Debug Subcommands (`/nc debug <system>`)

Each debug subcommand prints a live diagnostic report for that subsystem. Use `/nc debug all` for a full system dump.

<code>/nc debug radiation</code>	<code>/nc debug zombies</code>	<code>/nc debug ore</code>
<code>/nc debug smelter</code>	<code>/nc debug forge</code>	<code>/nc debug farming</code>
<code>/nc debug combat</code>	<code>/nc debug titan</code>	<code>/nc debug performance</code>
<code>/nc debug memory</code>	<code>/nc debug all</code>	

5. Commands — Admin System

Server Administration (nuclearcraft.admin)

`/nc health`

Live health snapshot: TPS, heap memory, machine count, titan status, database OK

Permission: nuclearcraft.admin

`/nc diagnostics`

Full startup checks + config validation suite — shows pass/fail for every check

Permission: nuclearcraft.admin

`/nc performance`

Detailed metrics: TPS, tracked objects, heap %, GC collections, machine load

Permission: nuclearcraft.admin

`/nc cleanup`

Purge all internal caches and request JVM garbage collection

Permission: nuclearcraft.admin

`/nc fixdata`

Validate all player data + configs; auto-repairs values that are out of range

Permission: nuclearcraft.admin

`/nc dumpdata`

Write full debug snapshot to plugins/NuclearCraft/dumps/<timestamp>.txt

Permission: nuclearcraft.admin

`/nc serverreport`

Print production summary: version, uptime, TPS, machines, validation results

Permission: nuclearcraft.admin

Radiation Admin (nuclearcraft.admin.radiation)

`/nc radiation check [player]`

Show radiation level, stage, contagion flag, and immunity timer

Permission: nuclearcraft.admin.radiation

`/nc radiation add <player> <n>`

Add N radiation points to a player (scale 0–1000)

Permission: nuclearcraft.admin.radiation

`/nc radiation remove <player> <n>`

Remove N radiation points from a player

Permission: nuclearcraft.admin.radiation

`/nc radiation set <player> <n>`

Set radiation to an exact value

Permission: nuclearcraft.admin.radiation

`/nc radiation clear <player>`

Remove all radiation instantly

Permission: nuclearcraft.admin.radiation

Zombie Admin (nuclearcraft.admin.zombies)

/nc zombie spawn irradiated

Spawn an Irradiated Zombie at your location

Permission: nuclearcraft.admin.zombies

/nc zombie spawn alpha

Spawn an Alpha Zombie (boss-tier, higher stats)

Permission: nuclearcraft.admin.zombies

/nc zombie stats

Show totals: spawned, active, alpha count

Permission: nuclearcraft.admin.zombies

/nc zombie surge start

Force-start a Radiation Surge event (mass zombie wave)

Permission: nuclearcraft.admin.zombies

/nc zombie surge stop

Force-stop the active Radiation Surge

Permission: nuclearcraft.admin.zombies

5. Commands — Machines & World

Ore Admin (nuclearcraft.admin.ore)

/nc ore spawn plutonium

Spawn a Plutonium Ore block at your location

Permission: nuclearcraft.admin.ore

/nc ore give fragment [n]

Give Raw Plutonium Fragments to yourself

Permission: nuclearcraft.admin.ore

/nc ore give drill

Give a Radiation Drill (required to mine ore safely)

Permission: nuclearcraft.admin.ore

/nc ore stats

Show tracked ore count and mining statistics

Permission: nuclearcraft.admin.ore

Smelter Admin (nuclearcraft.admin.smelter)

/nc smelter give [player]

Give a Nuclear Smelter block to yourself or a player

Permission: nuclearcraft.admin.smelter

/nc smelter stats

Show active machine count and aggregate processing stats

Permission: nuclearcraft.admin.smelter

/nc smelter debug

List every active smelter: location, state, progress %

Permission: nuclearcraft.admin.smelter

Forge Admin (nuclearcraft.admin.forge)

/nc forge give [player]

Give a Nuclear Forge block

Permission: nuclearcraft.admin.forge

/nc forge energy set <n>

Set energy of the forge you are looking at

Permission: nuclearcraft.admin.forge

/nc forge energy add <n>

Add energy to the forge you are looking at

Permission: nuclearcraft.admin.forge

/nc forge energy clear

Remove all energy from the forge you are looking at

Permission: nuclearcraft.admin.forge

/nc forge upgrade <mk1-mk4>

Force-upgrade the targeted item to a specific tier

Permission: nuclearcraft.admin.forge

`/nc forge stats`

Successes, failures, MK4 crafts, overload count

Permission: nuclearcraft.admin.forge

Equipment Admin (nuclearcraft.admin.equipment)

`/nc equipment give <type> [player]`

Give Plutonium or Hazmat equipment.

Types: plutonium-sword, plutonium-axe, plutonium-pickaxe, plutonium-shovel, plutonium-hoe, plutonium-helmet, plutonium-chestplate, plutonium-leggings, plutonium-boots,

`/nc equipment stats`

Show equipment worn by all online players

Permission: nuclearcraft.admin.equipment

Farming Admin (nuclearcraft.admin.farming)

`/nc farming give mutated-seed [n]`

Give Mutated Seeds

Permission: nuclearcraft.admin.farming

`/nc farming give healing-petal [n]`

Give Healing Petals (antidote component)

Permission: nuclearcraft.admin.farming

`/nc farming give radiation-antidote [n]`

Give Radiation Antidote (removes ~250 radiation)

Permission: nuclearcraft.admin.farming

`/nc farming give radiation-serum [n]`

Give Radiation Serum (120s radiation immunity)

Permission: nuclearcraft.admin.farming

`/nc farming growall [radius]`

Force-grow all mutated crops within radius (default 10)

Permission: nuclearcraft.admin.farming

`/nc farming stats`

Show active crops, bloom count, harvest statistics

Permission: nuclearcraft.admin.farming

5. Commands — Titan Boss & Titan Tech

Titan Boss Admin (nuclearcraft.admin.titan)

/nc titan spawn

Spawn the Plutonium Titan at your location (one Titan active at a time)

Permission: nuclearcraft.admin.titan

/nc titan kill

Instantly remove the active Titan (no loot drops)

Permission: nuclearcraft.admin.titan

/nc titan phase <1-4>

Force active Titan to a phase (1=Dormant, 2=Enraged, 3=Devastator, 4=Final)

Permission: nuclearcraft.admin.titan

/nc titan stats [player]

Titan history: summons, kills, deaths, damage dealt/taken, catastrophes survived, cores

Permission: nuclearcraft.admin.titan

Titan Technology Admin (nuclearcraft.admin.titan)

/nc titantech give <item> [n]

Give Titan Tech items. Valid types:

titan-reactor, forge, titan-helmet, titan-chestplate, titan-leggings, titan-boots, titan-sword, titan-axe, titan-pickaxe, titan-shovel, titan-hoe,

/nc titantech stats [player]

Show Titan Tech stats: gear crafted, radiation reflected, sword hits, arrows fired, full set active

Permission: nuclearcraft.admin.titan

/nc titantech aura

Display Titan weapon aura radius and trigger interval

Permission: nuclearcraft.admin.titan

/nc titantech setbonusinfo

List all active set bonus effects (requires full Titan armor set)

Permission: nuclearcraft.admin.titan

Combat Admin (nuclearcraft.admin.combat)

/nc combat stats [player]

Radiation kills, PvP hits, infections, total melee hits

Permission: nuclearcraft.admin.combat

/nc combat mastery [player]

Weapon mastery progress per type: sword, axe, bow, trident

Permission: nuclearcraft.admin.combat

/nc combat radiation [player]

Radiation combat stats: arrow hits, aura damage dealt

Permission: nuclearcraft.admin.combat

Full Titan Armor Set Bonuses:

- Radiation Immunity — complete immunity while all 4 pieces are worn
- Speed II + Jump Boost II — enhanced movement
- Resistance + Fire Resistance — reduced physical and fire damage
- +12 Max HP bonus (chestplate +4, full set +8 = 6 extra hearts)

6. Radiation System

Radiation Stages

All players start at 0 (Healthy). Radiation accumulates from sources below and decays slowly when not exposed. Scale: 0 to 1000.

#	Stage	Range	Effects
0	Healthy	0	No effects
1	Minor Exposure	1–249	Weakness I, mild nausea
2	Moderate Exposure	250–499	Weakness II, Slowness I, Hunger
3	Severe Exposure	500–749	Weakness III, Slowness II, Nausea, periodic damage
4	Critical Poisoning	750–1000	All above + rapid health drain + contagion spread

Radiation Sources

- % Mining Plutonium Ore without a Radiation Drill
- % Being hit by Irradiated Zombies or Alpha Zombies
- % Standing near an active Nuclear Smelter (machine aura)
- % Entering a Radiation Cloud spawned during zombie events
- % Being hit by another player's Plutonium weapon aura (PvP)
- % Being struck by Plutonium Arrows
- % Entering a Radiation Night event zone
- % Stage 4 contagion — nearby Stage 0–2 players gain radiation passively

Curing Radiation

Item / Method	Effect
Radiation Antidote	Removes ~250 radiation. Crafted from Healing Petals. Drops from mutated plants.
Radiation Serum	Grants 120s full immunity. Crafted from Antidotes + Irradiated Hearts.
Hazmat Armor (partial)	Each piece reduces radiation gain rate passively.
Titan Armor (full set)	Complete radiation immunity while all 4 pieces are worn.
Natural Decay	Radiation decays slowly over time when not exposed to sources.

7. Machines Guide

Nuclear Smelter

Processes Raw Plutonium Fragments into Refined Plutonium Ingots. Place like a furnace. Right-click to open GUI. Requires fuel to operate.

- % Input: Raw Plutonium Fragment or Plutonium Ore
- % Fuel: Coal, Charcoal, Blaze Rod, Lava Bucket, or any burnable
- % Output: Refined Plutonium Ingot
- % Overheat: prolonged use causes temporary shutdown — let it cool between batches
- % Machine emits a radiation aura — wear hazmat armor while near a running smelter
- % Machines in unloaded chunks are suspended and resume when the chunk loads

Nuclear Forge

Upgrades Plutonium equipment through MK I–IV tiers using Radioactive Cores. Higher tiers = better stats + higher risk.

Tier	Notes
MK I	Low failure risk. Low core cost. Safe starting tier.
MK II	Moderate failure risk. More cores required.
MK III	High failure risk. Item may be destroyed on failure.
MK IV	Maximum tier. Very high failure risk. Powerful stats on success.

Titan Reactor Forge

End-game machine for crafting Titan Technology items. Uses Titan Cores and Titan Fragments obtained by killing the Plutonium Titan. Produces Titan Armor, weapons and tools with unique abilities. Give it with:

```
/nc titantech give titan-reactor-forge
```

If your server crashes unexpectedly, run /nc fixdata on next startup to repair any orphaned machine states left by the unclean shutdown.

Machine Admin Tips

- '& Use /nc smelter debug to list all active smelters with state and progress %
- '& Place multiple smelters in parallel for higher throughput on large servers
- '& Machines drop as usable items when broken — custom data is preserved
- '& Use /nc forge stats to monitor upgrade successes, failures, and MK4 crafts
- '& The radiation aura is automatically throttled when server TPS drops below threshold

8. Plutonium Titan Boss

Overview

The Plutonium Titan is a custom multi-phase boss with unique AI, bossbar, AOE abilities, and proportional loot distribution. One Titan active at a time across all worlds. Recommended: 5–50 players per encounter.

Phases

Phase 1 — Dormant

100%–60% HP

Standard melee, periodic radiation bursts. Summons Irradiated Zombie waves.

Phase 2 — Enraged

60%–30% HP

Speed + damage boost. Radiation Slam (AOE) unlocked. Larger zombie waves.

Phase 3 — Devastator

30%–10% HP

All Phase 2. Radiation Beam attack. Triggers Radiation Night in arena area.

Phase 4 — Final

Below 10% HP

All abilities at reduced cooldowns, massive stat boost. Death = large loot drop.

Summoning

The Titan spawns automatically via server events (configurable in titan.yml). Admins can also manually spawn it:

```
/nc titan spawn      # Spawns at your location
/nc titan phase 2    # Jump to Enraged phase for testing
/nc titan kill        # Remove without loot (testing)
```

Loot

Loot is proportional to damage dealt. High-damage players receive rarer drops. All participants receive at least a consolation reward. Configure loot in titan_items.yml.

Titan Items (via Titan Reactor Forge)

Item	Description
titan-reactor-forge	Craft all Titan items from Cores + Fragments
titan-helmet/chestplate/leggings/boots	Full Titan armor set with all set bonuses
titan-sword/axe/pickaxe/shovel/hoe	Titan weapons with radiation aura on hit
titan-bow + titan-arrow	Ranged weapon firing plutonium arrows
titan-fragment	Raw crafting material dropped by the Titan
titan-core	Rare core used for highest-tier recipes

9. Troubleshooting

Expected Console Output on Clean Startup

```
[NuclearCraft] =====  
[NuclearCraft]   NuclearCraft: Plutonium Age v1.0.0  
[NuclearCraft]   Phase 14 Production – All systems active  
[NuclearCraft] =====  
[TestingManager]   All 6 startup checks PASSED.  
[ValidationManager] All 12 validations passed.  
[RecoveryManager]  No orphaned records. Clean shutdown confirmed.  
[ReleaseManager]   Java 21 '   ' | Heap 4096MB '   ' | Paper 1.21.4 '   '
```

Common Problems & Solutions

Plugin fails to load

Ensure Java 21+ and Paper/Purpur 1.21+. Check console for the exact error. Another plugin may conflict with /nc — check command aliases.

Database connection refused

Verify credentials in config.yml. Confirm MySQL is running, the database exists, and firewall allows the connection.

Items show as vanilla textures

Resource pack is not configured or player declined it. Verify URL and hash in resourcepack.yml. Set required: true to enforce.

Low TPS with many players

Run /nc performance to identify bottlenecks. Reduce particle-throttle-percent. Limit active smelters per chunk. Upgrade hardware.

Memory warnings in console

Run /nc cleanup to flush caches and request GC. If persistent, increase -Xmx (e.g. -Xmx4G). Run /nc dumpdata for analysis.

Radiation not working

Confirm the world is not excluded in config.yml worlds whitelist. Run /nc debug radiation for live system state.

Machines not saving on restart

Only /stop cleanly shuts down machines. After any crash, run /nc fixdata on next boot.

Bossbar not visible

Player must be within the Titan's bossbar range (titan.yml). Run /nc debug titan to check active boss state.

Config changes not applying

Run /nc reload after any edit. For database changes you must fully restart the server.

10. Quick Reference Card

Copy-Paste Admin Commands

Check server health <code>/nc health</code>	Run full diagnostics <code>/nc diagnostics</code>
View performance metrics <code>/nc performance</code>	Reload all configs <code>/nc reload</code>
Flush caches / request GC <code>/nc cleanup</code>	Repair corrupt player data <code>/nc fixdata</code>
Write debug dump to disk <code>/nc dumpdata</code>	Print production report <code>/nc serverreport</code>
Toggle debug logging <code>/nc debug</code>	Full debug dump all systems <code>/nc debug all</code>
Memory / heap diagnostic <code>/nc debug memory</code>	Check player radiation <code>/nc radiation check <player></code>
Clear player radiation <code>/nc radiation clear <player></code>	Set player radiation to 0 <code>/nc radiation set <player> 0</code>
Spawn Titan boss <code>/nc titan spawn</code>	Force Titan to final phase <code>/nc titan phase 4</code>
Remove Titan (no loot) <code>/nc titan kill</code>	Give Titan chestplate <code>/nc titantech give titan-chestplate</code>
Spawn irradiated zombie <code>/nc zombie spawn irradiated</code>	Start radiation surge <code>/nc zombie surge start</code>
Stop radiation surge <code>/nc zombie surge stop</code>	Give radiation antidote x10 <code>/nc farming give radiation-antidote 10</code>
Give radiation serum x5 <code>/nc farming give radiation-serum 5</code>	Give radiation drill <code>/nc ore give drill</code>
Give nuclear smelter <code>/nc smelter give</code>	Give nuclear forge <code>/nc forge give</code>

Config Files at a Glance

config.yml Main: database, performance, worlds, admin	radiation.yml Stages, damage/tick, decay, contagion range
titan.yml Boss HP phases, ability timing, bossbar range	titan_items.yml Boss loot table and drop weights
smelter.yml Recipes, fuel values, overheat threshold	forge.yml Upgrade tiers, energy cost, success/fail rates
farming.yml Crop growth, mutation chance, antidote recipes	zombies.yml Spawn rates, alpha chance, surge config
ore.yml Vein size, depth, rarity per world	messages.yml All player-facing strings (localisable)
resourcepack.yml Pack URL, SHA-1 hash, required flag	balance.yml Economy rewards, XP rates, loot multipliers

